a cyberwitchery talk

ai architecture katas

learning by building small models in plain python

veit heller
cyberwitchery lab

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section 01

01 why tiny models?

you (probably) have already used one

- a chatbot, a translator, code completion.
- under each is a neural network: arrays of parameters and matrix multiplication.
- you could build one.

frameworks are for production code

for learning, we will resort to minimalism.

the practice

build a small, runnable version yourself.

mostly python and numpy and small enough to read fully.

what you trade

+ visibility: every intermediate value is in front of you.
+ debuggability: a wrong value traces to one line.
+ intuition: the shapes show what each piece is for (puzzles, y'all!).

| out of scope: scale, throughput, benchmark numbers.

the setup

- just **numpy arrays**, a loop, maybe a plot.
- **shape-first**: the shapes are most of the program.
- the notation in papers is shorthand for the array code.

one trip to the terminal to build a transformer ourselves.

section 02

02 a tiny transformer, end to end

what a transformer computes

tokens
→ embeddings
→ (attn + mlp) ×N
→ outputs

a stack of identical layers. each updates every position from all positions.

a token is a chunk of text, an embedding its vector. attention mixes across positions, the mlp transforms each position on its own.

self-attention

$$\text{scores} = Q\,K^{\top} / \sqrt{d_k}$$

$$\text{out} = \text{softmax}(\text{scores})\,V$$

a weighted average of all positions, with data-dependent weights.

Q, K, V are linear projections of the input. scores is (T, T): how much each position draws from the others.

softmax

```
def softmax(x):
    x = x - x.max(-1, keepdims=True)   # row max
    e = np.exp(x)
    return e / e.sum(-1, keepdims=True)
```

softmax maps a vector to a probability distribution. subtracting the row max is the numerically stable form: identical result, no overflow in exp.

residual connections

```
x = x + attn(ln(x))
x = x + mlp(ln(x))
```

the addition lets gradients reach every layer.

the residual connection, from resnets (2015). deep stacks need it to train.

layer normalization

$$y = \gamma \ (x - \mu) \ / \ \sqrt{\sigma^2 + \varepsilon} \ + \beta$$

keeps activations from drifting as depth grows.

μ, σ: mean and variance over the feature axis, per position. γ, β: learned. ε: avoids dividing by zero.

→ live

let’s build one.

invariants need asserts

- each row of `softmax(scores)` sums to **`1`**.
- shape is preserved: `(T, d)` $\rightarrow$ `(T, d)`.
- layer norm output has **`mean 0, variance 1`** before γ, β.
- permute the positions, and the outputs permute identically.

a transformer block

~60 lines of numpy, no framework

attention, layer normalization, an mlp, residual connections, and the tests that pin them down. production models are this block, only wider and deeper. but: the operations are the same!

section 03

03 paper to prototype

read an equation as an array program

- every symbol is an array, every subscript is an axis.
- a $\sum_j$ is a matmul or a `.sum(axis=j)`.
- write the **shapes** in a comment before the code.

if the shapes line up, you usually understand the mechanics of the formula.

catch bugs early with boring things

- `shape asserts` at every boundary, not just the end.
- `unit tests for the math`: a known input, a hand-checked output.
- the `overfit test`: fit ten examples exactly. if you can't, it's broken.

section 04

04 multi-scale modelling (rgm)

structure across scales

many signals carry structure at several scales at once:

images, audio, text, physical fields.

model the coarse structure first, then refine it.

coarse to fine

$$\text{down: } (N) \rightarrow (N/2)$$
$$\text{up: } (N/2) \rightarrow (N)$$

a resolution pyramid: model coarse structure first, detail after.

coarse levels are small and global, fine levels carry details.

renormalization

$$x_\ell = \texttt{block}(\texttt{down}(x_{\ell+1}))$$

for each level ℓ

apply the *same* operator at every scale.

borrowed from the renormalization group in physics: coarse-grain, model, repeat. `block` is the same weights at every level ℓ.

generativity

run the pyramid the other way: start coarse,
invent detail scale by scale.

a learned model at each scale adds the next level of detail. image example: it produces new images, it doesn't just process the ones you give it.

section 05

05 state-space models

attention has a cost problem

it compares every pair of positions, so doubling the input quadruples the work.

state-space models do sequence modelling in linear time: audio, dna, whole documents.

a state-space model

a loop with memory: walk the sequence, carry a running state, update it each step.

`state = A state + B input`, then `output = C state`. like an rnn, but every step is linear.

one model with two forms

because the loop is linear, it is also a single convolution over the sequence.

train as the convolution (whole sequence in parallel), run as the recurrence (one step, cheap memory). using the same weights!

attention also has a memory problem

to make the next token, it holds every earlier token in memory.

that store grows with the sequence and never shrinks.

constant memory

a fixed-size state summarises the whole past,

however long the sequence.

and it's durable: save or reload it. state is a first-class object.

section 06

06 your own tiny model lab

a reusable harness

we looked at three very different models.

we used the same scaffold around each.

a dataset you can overfit, a shape-checked forward pass, invariants as tests. build infra for your lab!

when to stop

the toy is for understanding.

when you need scale, reach for pytorch, jax, w/e.

the takeaway

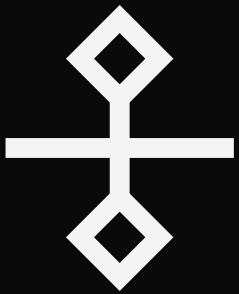
a paper is a specification.

implement the smallest version that runs.

try to stay true to the ideas only.

thanks.

questions?



veit heller veit@veitheller.de

slides: github.com/hellerve/talks